FULL NAME: ______________________________ **BLOCK:** ____________

Logarithmic Functions + DOK-3 Sales Revenue Inverse (Item 28)

Lesson 6-4 · Quick · Teacher Edition

OBJECTIVES

Math Objective	Graph logarithmic functions, identify key features (asymptote, domain, range, intercepts), and find equations of inverses of exponential and logarithmic functions.
Language Objective	Describe how transforming a log function shifts its asymptote and key points; explain why the inverse of a log is an exponential.
Essential Understanding	Log and exponential functions are inverses — swap x and y, solve. The asymptote of a log is the y-axis mirror of an exp function's horizontal asymptote.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency with graphing log functions and finding inverses; apply to the Sales Revenue DOK-3 task (Item 28 rehearsal for Form B model-then-invert structure).
Essential Question	How do the graphs of logarithmic and exponential functions relate, and how do we find one from the other?
Materials	Student packet, pencil, calculator. Rules card: $\log_b 1 = 0$; $\log_b b = 1$; $y = \log_b x$ has VA at $x = 0$; $y = \log_b(x - h) + k$ has VA at $x = h$; inverse of $y = a \cdot b^x$ is $y = \log_b(x/a)$; inverse of $y = \log_b(x - h) + k$ is $y = b^{(x-k)} + h$.

[PIN] PRE-FLIGHT — SINGLE-PERIOD COMPRESSED LESSON + DOK-3 CAPSTONE

This is a single-period quick treatment of Lesson 6-4. ONE example modeled in Launch (Ex 3B: inverse of $g(x) = \log_7(x + 5)$ + composition verification). Contrast with Ex 3A (inverse of exponential) in 90 sec only — same swap-and-solve process, direction reversed. Release to Explore at minute 15.

DOK-3 driver: Practice #28 ($R = 12\log(a + 1) + 25$). Students (a) find the inverse function $a = 10^{\frac{R-25}{12}} - 1$, (b) judge which form is easier for specific R values. Real Savvas values from SE source.

Pace: [STAR] q28 (10 min) → q21 (4 min) → q24 (4 min) → q13 (4 min) → q9 (3 min). ∼25 min total Explore.

45-min Wed F variant: cut q9. Never cut q28.

Assessment alignment: q21→Form B Q7, q24→Q12, q13→Q11, q9→Q9/Q10, q28→DOK-3 rehearsal.

Tag: bridge to log inverses launch

Do Now **DOK:** 1–2 **Minutes:** 7

Teacher does

- Hand out at door.
- Circulate 4 min.
- Cold-call on q31: ask which transformation shifts the VA.
- Pull sheets at minute 7.

Students do

- q31: translation multi-select for $h(x) = \ln(x+2) - 1$.
- q5: graph $y = \log_4 x$ with all key features.
- q32: find inverse of $f(x) = 5^{x+1}$ (SAT/ACT).

Questions to ask

- q31: how does $(x+2)$ inside the log change the VA?
- q5: domain? VA? x-intercept? End behavior as $x \to 0^+$ and $x \to \infty$?
- q32: undo 5^x using log base 5 — what is the inverse operation?

Adult role
Solo teacher: circulate silently. No hints on q32.

Do Now 1 — Translation identification: $h(x) = \ln(x+2) - 1$

The logarithmic function $g(x) = \ln x$ is transformed to $h(x) = \ln(x+2) - 1$. Which of the following are true? Select all that apply.

☐ $g(x)$ is translated 2 units upward.

☐ $g(x)$ is translated 2 units to the right.

☐ $g(x)$ is translated 2 units to the left.

☐ $g(x)$ is translated 1 unit downward.

☐ $g(x)$ is translated 1 unit to the left.

☐ The vertical asymptote shifts 2 units to the left.

☐ The vertical asymptote shifts 2 units to the right.

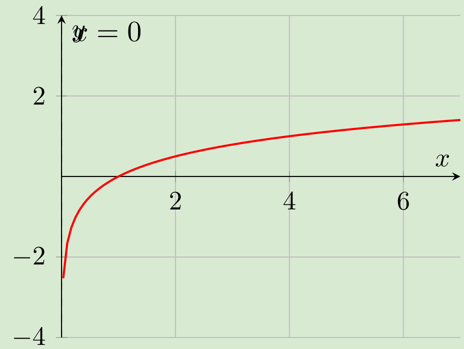
[CHECK] ANSWER KEY

Answer key (from source): $g(x)$ is translated 2 units to the left; $g(x)$ is translated 1 unit downward; The vertical asymptote shifts 2 units to the left.

Do Now 2 — Graph $y = \log_4 x$; state key features

Graph the function $y = \log_4 x$ and identify the domain and range. List any intercepts or asymptotes. Describe the end behavior.

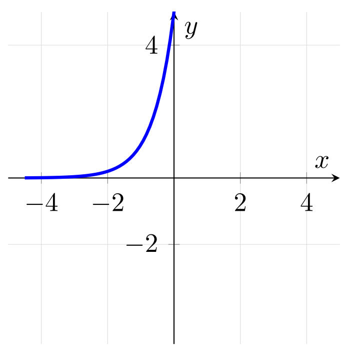
[CHECK] ANSWER KEY



Answer key (from source): Domain: $\{x \mid x > 0\}$; range: all real numbers; x-intercept: 1; asymptote: y-axis; end behavior: As $x \to 0$, $y \to -\infty$. As $x \to \infty$, $y \to \infty$.

Do Now 3 — SAT/ACT: inverse of $f(x) = 5^{x+1}$

SAT/ACT The graph shows the exponential function $f(x) = 5^{x+1}$. Which of the following functions represents its inverse, $f^{-1}(x)$?



Ⓐ $f^{-1}(x) = 1 + \log_5 x$ Ⓑ $f^{-1}(x) = \log_5 x - 1$
Ⓒ $f^{-1}(x) = \log_5(x - 1)$ Ⓓ $f^{-1}(x) = \log_5(x + 1)$

[CHECK] ANSWER KEY

Answer key: Ⓑ $f^{-1}(x) = \log_5 x - 1$. Swap x and y: $x = 5^{y+1}$, so $y + 1 = \log_5 x$ and $y = \log_5 x - 1$.

[CHECK] ANSWER KEY

Answer key (from source): $g(x)$ is translated 2 units to the left; $g(x)$ is translated 1 unit downward; The vertical asymptote shifts 2 units to the left.
(verify before class)
(verify before class)

Tag: teacher models Ex 3B (inverse of log) + 90s Ex 3A contrast

Launch **DOK:** 2 **Minutes:** 8

Teacher does

- Project Ex 3B. Think aloud: $g(x) = \log_7(x+5)$. Step 1: swap x and y. Step 2: write $7^x = y+5$. Step 3: solve — $y = 7^x - 5$.
- "Now verify with composition: $g(g^{-1}(x)) = \log_7((7^x - 5) + 5) = \log_7(7^x) = x$. ✓"
- 90-sec contrast: Ex 3A (inverse of exponential, e.g. $y = 3 \cdot 5^x$). "Swap x and $y \to x = 3 \cdot 5^y \to x/3 = 5^y \to y = \log_5(x/3)$. Same swap-and-solve."
- Flag the Rules card — students copy VA shift rule and inverse rules into margin.
- 30-sec pause: partner repeats the inverse steps in their own words.
- Do NOT solve Explore items — release at minute 15.

Students do

- Watch Ex 3B silently. Note swap-then-solve sequence.
- Copy composition verification step.
- During 90-sec contrast: note that inverting an exponential gives a log, and inverting a log gives an exponential.
- Copy Rules card VA and inverse rules into margin.

Questions to ask

- In $g(x) = \log_7(x+5)$, what is the domain? VA? ($x = -5$)
- After swapping x and y, how do we isolate y when $7^x = y + 5$?
- What is $g^{-1}(x)$? What is its domain? ($x \in \mathbb{R}$)
- Why does the VA of g become the HA of g^{-1}?

Adult role

Project Ex 3B. Model composition check. 90-sec Ex 3A contrast. Release promptly at minute 15.

Example 3B

Inverses of Exponential and Logarithmic Functions

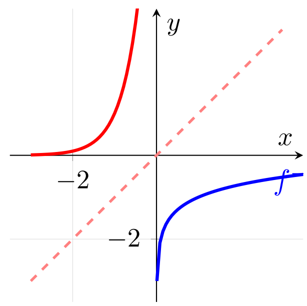
What is the equation of the inverse of the functions?

A. $f(x) = 10^{x+1}$

$$\begin{aligned} y &= 10^{x+1} \\ x &= 10^{y+1} \quad \text{(Rewrite the function as } y =\text{, interchange } x \text{ and } y\text{, and solve for } y.\text{)} \\ y + 1 &= \log x \\ y &= \log x - 1 \end{aligned}$$

The equation of the inverse of $f(x) = 10^{x+1}$ is $f^{-1}(x) = \log x - 1$.
Verify by function composition.

$$\begin{aligned} f(f^{-1}(x)) &= 10^{(\log x - 1)+1} \\ &= 10^{\log x} \\ &= x \end{aligned} \qquad\qquad \begin{aligned} f^{-1}(f(x)) &= \log 10^{x+1} - 1 \\ &= x + 1 - 1 \\ &= x \end{aligned}$$



CONCEPTUAL UNDERSTANDING / LOOK FOR RELATIONSHIPS

$f(x)$ is a translation of the parent function $y = 10^x$ one unit left. $f^{-1}(x)$ is a translation of the parent function $y = \log x$ one unit down. Graphical translations of a function and its inverse are directly related. Explain.

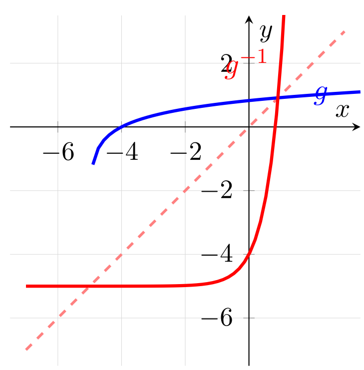
B. $g(x) = \log_7(x+5)$

$$\begin{aligned} y &= \log_7(x+5) \quad \text{(Rewrite as } y =.) \\ x &= \log_7(y+5) \\ y+5 &= 7^x \\ y &= 7^x - 5 \end{aligned}$$

The equation of the inverse of $g(x) = \log_7(x+5)$ is $g^{-1}(x) = 7^x - 5$.
Verify by function composition.

$$\begin{aligned} g(g^{-1}(x)) &= \log_7(7^x - 5 + 5) \\ &= \log_7 7^x \\ &= x \end{aligned}$$

$$\begin{aligned} g^{-1}(g(x)) &= 7^{\log_7(x+5)} - 5 \\ &= x + 5 - 5 \\ &= x \end{aligned}$$



[CHECK] ANSWER KEY

The equation of the inverse of $f(x) = 10^{x+1}$ is $f^{-1}(x) = \log x - 1$.
The equation of the inverse of $g(x) = \log_7(x+5)$ is $g^{-1}(x) = 7^x - 5$.

[MIC] TEACHER SCRIPT

1. “Watch me find the inverse of a LOG function.”
2. “Step 1: replace $g(x)$ with y. Step 2: swap x and y.”

3. "Step 3: rewrite in exponential form — $7^x = y + 5$."
4. "Step 4: solve for y — $y = 7^x - 5$. That's $g^{-1}(x)$."
5. "Composition check: $g(g^{-1}(x)) = \log_7(7^x - 5 + 5) = x$. ✓"
6. "Now 90 seconds: Ex 3A — inverting an exponential gives a log. Same swap-and-solve."
7. Release to Explore.

Tag: TPS + DOK-3 driver

Explore **DOK:** 2–3 **Minutes:** 25

Teacher does

- 4 laps across 25 min.
- Lap 1 (10 min): [STAR] q28 DOK-3. Press pairs on inverse setup and form comparison. DO NOT give the inverse formula — redirect with 'what does it mean to undo R?'
- Lap 2 (4 min): q21. Press notation: what is the inverse of 5^{x-3}?
- Lap 3 (4 min): q24. Press: inverse of $\log_2(8x)$ — watch for domain.
- Lap 4 (4 min): q13. Press: how does the horizontal shift move the VA?
- Lap 5 (3 min): q9. Are graphs inverses? Key test: reflective symmetry over $y = x$.
- 45-min Wed F: skip q9.

Students do

- 5 items in order: [STAR] q28 → q21 → q24 → q13 → q9.
- For q28: BEFORE computing, write down $R = 12\log(a + 1) + 25$ and plan how to isolate a.
- For q28 part 2: decide which form (original or inverse) is easier for $R = 49$ and $R = 61$. Write a justification sentence.

Questions to ask

- q28: how do you undo $+25$ first? Then undo $\times 12$?
- q28: after $(R - 25)/12 = \log(a + 1)$, what is the next step?
- q28: for $R = 49$, would you use $R = 12\log(a + 1) + 25$ or $a = 10^{\frac{R-25}{12}} - 1$? Why?
- q21: identify base and exponent — what is the log that undoes 5^x?
- q24: inverse of $\log_2(8x)$ — swap x and y first. Then rewrite.
- q13: find the VA of the translated log from the graph.
- q9: if two graphs are inverses, what visual test do they pass?

Adult role

Solo teacher: 4 laps. On q28, press the judgment step (which form and why) — that is the DOK-3 performance criterion.

[SPEAK] CHAIN 1 CALLBACK — Revenue Inverse (PEAK, 5 priors)

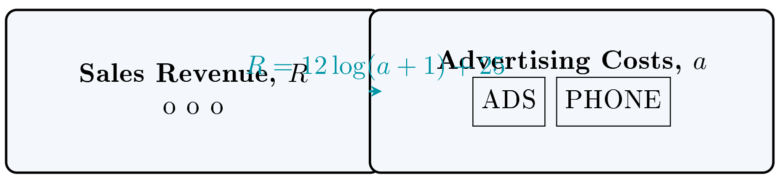
When to say: Before students start Practice #28.

"We've done this five times now: Storage Box, wavelength, cylinder, Richter, and today revenue. Every time: a model, a constraint, a hidden input we want to recover. Today's twist: the original model is **exponential**, so to invert it we need a **logarithm**. That's why we spent yesterday on logs. A log isn't a new topic — it's the specific tool that lets us invert an exponential model. Same job we've been doing all year, new lock, new key."

Note: q33 (Telescope Magnitude, fast-finisher) is the Chain 1 micro-callback: "Another physical formula, another extraction. At this point you should be able to see the shape from a mile away — constraint plus inversion."

Practice #28 [STAR] DOK-3 SALES REVENUE INVERSE — Form B model-then-invert rehearsal

A company uses this function to relate sales revenue, R, and advertising costs, a. What is the equation of the inverse of the equation? Which equation would be easier to use to find a value of a for a particular value of R?



[STAR] DOK-3 SALES REVENUE — Practice #28

Students must (a) isolate a from $R = 12\log(a+1) + 25$, yielding $a = 10^{\frac{R-25}{12}} - 1$, (b) judge which form is easier for specific R values.
Key criteria: (1) correct algebraic isolation steps shown, (2) inverse formula correct (base 10, not natural log), (3) justification for form preference cites a specific computation.
Assessment alignment: DOK-3 rehearsal for Form B model-then-invert structure

[CHECK] ANSWER / ANTICIPATED TALK

The inverse of the formula is $a = 10^{\frac{R-25}{12}} - 1$. It would be easier to use the inverse to find a when R is known.
Assessment alignment: DOK-3 rehearsal for Form B model-then-invert structure

Practice #21 — Inverse of exponential 5^{x-3} [Form B Q7 rehearsal]

Find the equation of the inverse of each function. $f(x) = 5^{x-3}$.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $f^{-1}(x) = \log_5 x + 3$
Assessment alignment: Form B Q7: inverse of $5^{(x-3)}$

Practice #24 — Inverse of $\log_2(8x)$, token drag [Form B Q12 rehearsal]

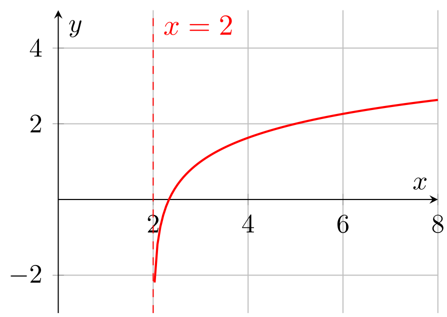
Find the equation of the inverse of each function. $f(x) = \log_2(8x)$.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $f^{-1}(x) = \frac{2^x}{8}$
Assessment alignment: Form B Q12: inverse of $\log_2(8x)$, token drag

Practice #13 — Asymptote of translated log from graph [Form B Q11 rehearsal]

Use Structure The graph shows a transformation of the parent graph $f(x) = \log_3 x$. Write an equation for the graph.

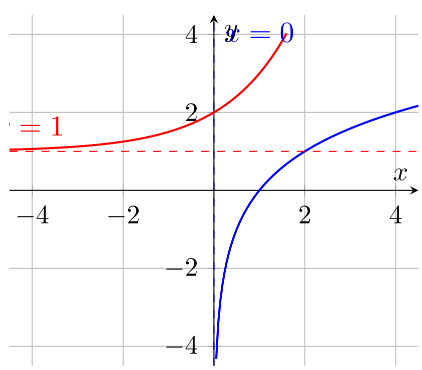


[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $g(x) = \log_3(x-2) + 1$
Assessment alignment: Form B Q11: asymptote of translated log from graph

Practice #9 — Are graphs inverses? Inverse from graph [Form B Q9/Q10 rehearsal]

Look for Relationships Are the logarithmic and exponential functions shown inverses of each other? Explain.



[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): No; The x-intercept of the blue function is not equal to the y-intercept of the red function.
Assessment alignment: Form B Q9/Q10: are graphs inverses; inverse from graph

Tag: academic conversation + 6-5 bridge

Share / Summary **DOK:** 2 **Minutes:** 5

Teacher does

- Cold-call 1 pair to present q28 solution and form-choice justification.
- Class signals thumbs. Write $a = 10^{\frac{R-25}{12}} - 1$ on board.
- 6-5 bridge: "Log properties (product/quotient/power rules) will let you expand and condense log expressions — making inverses and equations faster."

Students do

- Presenting pair: read inverse formula and cite which form is easier for $R = 49$ and $R = 61$.
- Rest of class: signal thumbs and write bridge note in margin.

Questions to ask

- How did you isolate a from $R = 12\log(a+1) + 25$?
- For $R = 49$, which form is computationally easier? Why?

Adult role

Cold-call a pair that struggled on the judgment step — hold them accountable to citing a computation, not just a preference.

Tag: inverse + asymptote (not graded)

Exit Ticket **DOK:** 1–2 **Minutes:** 5

Teacher does

- Collect at door.
- Scan: (1) Did students swap x/y correctly? (2) Did students solve for y in exponential form? (3) Did they state VA of f as $x = 2$ and HA of f^{-1} as $y = 2$?

Students do

- Two-part custom exit: find inverse, then state both asymptotes.

Questions to ask

- Inverse of $f(x) = 3\log_5(x-2) + 1$: swap → exponential form → solve.
- VA of f is $x =$ ___ (vertical asymptote of original log).
- HA of f^{-1} is $y =$ ___ (the VA of f becomes the HA of the inverse).

Adult role

Collect. No grade. Use responses to plan any re-teach before 6-5.

[CLIPBOARD] EXIT TICKET — Student Form

Find the inverse of $f(x) = 3\log_5(x-2) + 1$.
State the vertical asymptote of f.
State the horizontal asymptote of f^{-1}.
One biggest thing I learned today: ______________________________

ANSWER KEY (TE only): $f^{-1}(x) = 5^{\frac{x-1}{3}} + 2$. VA of f: $x = 2$. HA of f^{-1}: $y = 2$.

[CLOCK] PERIOD A / PERIOD F — TIMING NOTES

50-min base: standard pacing above.
45-min Wed F variant: drop q9 from Explore. Never cut q28.
If finished early: hand out Practice #33 (Telescope PT — stretch).

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Log Function Key Features + Inverse Rules card as a sticker/cut-out.
- For q28, provide the setup: $(R - 25)/12 = \log(a + 1)$. Students solve from that point.
- Accept verbal justification for form-choice in lieu of written sentence.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: 'vertical asymptote' = a line the graph never touches; 'inverse function' = reverses the input/output; 'composition' = plugging one function inside another.
- 'log base b of x' = the exponent you put on b to get x.
- Pair ELL students with English-dominant partners for TPS.